Product Specifications

Item	DAQ-08	DAQ-04	DAQ-02
Electrical Specifications			
Electrical safety class	Class III		
Power supply	5 V DC via USB micro-B		
Maximum current consumption	0.9 A		
Maximum power consumption	4.5 W		
Signal Specifications			
Electronic hardware type	Field-Programmable Gate Array (FPGA)		
Input signal type	SMA: 2.5 V digital LVTTL at 50 Ω USB Type-C: LVDS		
Input polarity	positive		
Minimum input pulse width	> 1.5 ns		
Minimum input pulse height (abs)	2.5 V		
Maximum input pulse height (abs)	5 V		
Single-ended SMA inputs (LVTTL)	11	7	5
USB Type-C LVDS inputs	12	8	6
Laser trigger input (SMA, sync input)	1		
Laser trigger output (SMA, sync output)	1		
Output signal standard	SMA: 2.5 V digital LVTTL at 50 Ω USB Type-C: LVDS		
Impedance	50 Ω		
PC connection	USB micro-B port		
Timing variation (digital signal)	80 ps RMS		
Maximum repetition frequency	80 MHz		
Minimum resolvable fluorescence lifetime	50 ps		
Single-event precision ($\sigma/\sqrt{2}$)	300 ps		
Supported laser sync frequencies	Input: 10 MHz to 80 MHz Output: 10, 20, 40, 80 MHz		
Time bin width (temporal resolution)	48 ps @ 80 MHz 96 ps @ 40 MHz 192 ps @ 20 MHz 384 ps @ 10 MHz		

Item	DAQ-08	DAQ-04	DAQ-02
Dead time	1.5 ns		
Differential non-linearity	< 0.5% RMS		
Maximum transfer rate (sum over channels)	100 M counts/s		
Maximum burst rate per input channel	640 M counts/s		
Minimum pixel dwell time	100 ns		
Maximum duty cycle of reference signals	80 %		
Maximum acquisition duration	Not limited by the acquisition card		
Software Requirements			
Acquisition software	FLIM Labs *Spectroscopy Py* or *FLIM IMAGER*		
Operating system	Windows 10 or later		
PC hardware	multicore CPU, 4GB RAM		
Physical Specifications			
Dimensions (L x W x H)	139 mm x 101 mm x 28 mm		
Weight	120 g		
Protection rating	IP40		
Environmental Specifications			
Operating temperature	0°C to +40°C		
Storage temperature	-20°C to +70°C		
Relative humidity	Up to 80% non-condensing		